Name: ___________________________ Class: ____________ Date: ____________ **ID: A**

Chemistry Chapter 17 Practice Quiz 2 — Le Châtelier's Principle

Multiple Choice

Identify the choice that best completes the statement or answers the question.

___ 1. Le Châtelier's principle states that when a stress is placed on a reversible process, it will

a. stop the reaction completely
b. proceed in the direction that relieves the stress
c. always shift to the right
d. double the value of Keq

___ 2. If the concentration of a reactant is increased in a system at equilibrium, the equilibrium will shift

a. to the left, toward more reactants
b. to the right, toward more products
c. there is no shift — concentration changes don't affect equilibrium
d. in the direction that increases temperature

___ 3. For a gaseous reaction, increasing pressure causes the equilibrium to shift toward

a. the side with more moles of gas
b. the side with fewer moles of gas
c. the product side always
d. no shift — pressure never affects equilibrium

___ 4. Which of the following stresses DOES change the value of Keq?

a. Adding more reactant
b. Removing a product
c. Changing the temperature
d. Adding a catalyst

___ 5. The formation of ammonia ($N_2 + 3H_2 \rightleftharpoons 2NH_3$) is exothermic. Increasing the temperature will

a. shift equilibrium right and increase Keq
b. shift equilibrium left and decrease Keq
c. shift equilibrium right and decrease Keq
d. have no effect on Keq

True / False

Indicate whether the statement is true or false.

___ 6. Changing the concentration of a reactant or product shifts the equilibrium but does not change Keq.

___ 7. Increasing pressure always shifts a gaseous equilibrium to the right.

___ 8. Adding a catalyst changes both the equilibrium position and the value of Keq.

___ 9. If a product is removed from a system at equilibrium, the equilibrium shifts to the right.

___ 10. For an exothermic reaction, increasing the temperature is like adding more product.

___ 11. Decreasing pressure shifts a gaseous equilibrium toward the side with fewer moles of gas.

___ 12. Adding a solid reactant to a heterogeneous equilibrium shifts the equilibrium to the right.

___ 13. A catalyst helps a system reach equilibrium faster but does not change the Keq value.

___ 14. For an endothermic reaction, increasing temperature increases the Keq value.

___ 15. Removing a reactant from a system at equilibrium will cause the equilibrium to shift left.

Short Answer

16. Consider the exothermic reaction: $CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$ State the direction of the equilibrium shift (left or right) for each of the following stresses: a. Pressure is increased b. Temperature is increased c. CO is removed d. H_2O is added

17. Explain in your own words what Le Châtelier's principle means. How can chemical manufacturing industries use it to maximize the yield of a reaction?

18. A student claims: 'Adding a catalyst to a reaction at equilibrium will shift the equilibrium to the right because the forward reaction speeds up.' Is the student correct? Explain your answer fully.

Essay

Answer TWO of the following. You may answer all 3 for possible bonus.

19. Describe all three types of stresses that can be applied to a system at equilibrium: concentration, pressure, and temperature. For each, explain (a) how the equilibrium position is affected, and (b) whether the value of Keq changes. Give one example for each.

20. Explain fully how temperature changes affect a system in equilibrium. Why does changing temperature change the Keq value, while changing concentration or pressure does not? Use the concept of thermal energy as a reactant or product in your answer.

21. The Haber process ($N_2 + 3H_2 \rightleftharpoons 2NH_3$) is exothermic and is used industrially to produce ammonia. Using Le Châtelier's principle, explain what conditions of temperature, pressure, and

concentration a manufacturer would choose to maximize ammonia production, and describe the trade-offs involved.